



Loops: Code That Repeats

Ontario Math C3.1 — repeating events

Name:

Date:

★ I can write and change a repeat loop to make a sprite repeat actions.

What is a loop?

Sometimes you want your sprite to do the same thing again and again. You could drag the same block many times — but that makes a long, messy script.

A repeat loop is a block that does the repeating for you. You put the actions inside it once, and tell it how many times to run them. Bit will show you how.

Worked example — read this script

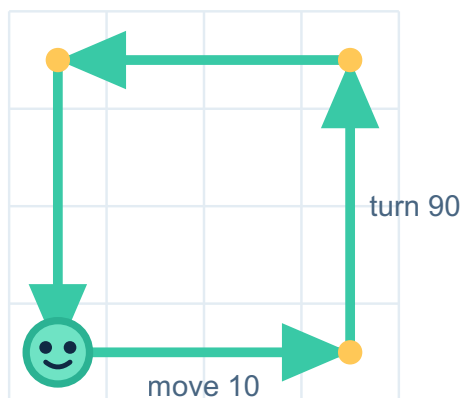
when green flag clicked

repeat 4

move 10 steps

turn 90 degrees

The two blocks inside repeat run 4 times. Move, turn — four times — draws a square and brings Bit back to the start.



Bit starts here

Bit draws a square: move 10, turn 90, four times.

The same idea in Python

PYTHON

```
for side in range(4):  
    bit.forward(10)  
    bit.right(90)
```

OUTPUT

Bit draws a square – 4 sides, back to the start.

1 Finish the loop (write)

Make Bit draw a TRIANGLE (3 sides). Fill in the two missing numbers: repeat ____, and turn ____ degrees. (Hint: a triangle's turns add up to 360.)

2 Change it, then predict (read & alter)

Anya changes repeat 4 to repeat 6, and turn 90 to turn 60. What shape does Bit draw now, and how many sides does it have? Write your answer and explain how you know.

3 Predict, then check

This loop has TWO move blocks inside it: repeat 3 { move 20 steps, move 20 steps }. How many steps does Bit move in total? Predict first, then run it to check.
